

Telecommunication

Why the telephone network needed to go digital end to end?

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The traditional Public Switched Telephone Network (PSTN) was designed mainly for voice communication. To support modern communication services, the telephone network needed to become digital end to end.

Reasons -

1. Analog local loop -

The subscriber line carried analog signals only.

2. Designed only for voice -

The PSTN was only designed for voice communication.

3. Low data rate -

Data transmission rate limited to 0.6-56 Kbps.

4. Noise & distortion -

Traditional PSTN was highly affected to noise.

5. Separate networks -

Classic PSTN used separate network for voice & data transfer.

6. Need of multiple digital services -

The growing demand for internet, video & file transfer required a digital network.

ISDN overcame all these issues by extending digital transmission all the way to the subscriber over the existing telephone network.

Define ISDN -

=> Integrated Services Digital Network (ISDN) is a set of communication standards that allows the digital transmission of voice, video, data and other services over traditional telephone networks.

Integration - Voice, data, video on one subscriber line and one number

End to End digital - Circuit switched 64 kbps channels with low noise and fast setup.

Standard interfaces - Uses a small set of ITU-T standard sockets.

Out of Band signalling - A separate D channel for call control

Explain ISDN channel -

=> ISDN uses 3 types of logical channels to transmit different types of information over the digital subscriber line -

1. B Channel
2. D Channel
3. H Channel

1. B channel -

Speed - 64 kbps

Purpose - Carries user information.

It can carry - Voice Video
Data

2. D channel -

Speed - 16 kbps in BRI

64 kbps in PRI

Purpose - Carries signalling and control information.

Used for - Dialing, calling setup, call ~~transmission~~ termination.

3. H channel -

Speed - H0 - 384 kbps

H11 - 1536 kbps

H12 - 1920 kbps

Purpose - Used for high-speed communication.

Used for - Video conferencing

Basic Rate Interface (BRI): 2B+D

One twisted pair carries 3 multiplexed channels -

B1 - 64 kbps (voice/data)

B2 - 64 kbps (voice/data)

D - 16 kbps (signalling)

$$\therefore \text{User Rate} = 2 \times 64 + 16 = 144 \text{ kbps}$$

$$\text{Line Rate} = 192 \text{ kbps } (144 + 48)$$

Primary Rate Interface - V

	North America/Japan(T1)	Europe/Bangladesh(E1)
Structure	23B + D	30B + D
B Channels	23 x 64 kbps	30 x 64 kbps
D Channels	64 kbps	64 kbps
Line rate	1.544 Mbps	2.048 Mbps
Typical use	PBX trunks, call centres	PBX trunks, ISPs, Video links.

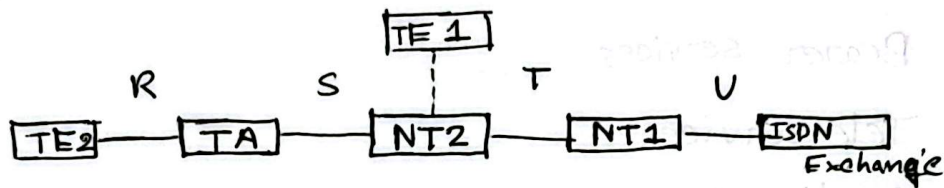
BRI vs PRI -

	BRI	PRI
Structure	→ 2B + D	23B + D (T1) 30B + D (E1)
B channel	→ 2 x 64 kbps	23 x 64 kbps 30 x 64 kbps
D channel	→ 16 kbps	64 kbps
Line Rate	→ 192 kbps	1.544 Mbps 2.048 Mbps
Application	→ Home & small office	Business, call center.

$$\text{Line Rate} = 192 \text{ kbps} = 2 + 16 \times 12 = 192 \text{ kbps}$$

$$(2 + 16 \times 12)$$

ISDN Architecture -



Functional groups -

1. TE 1 - Terminal Equipment 1 is a Native ISDN Device. (Digital phone, ISDN Port).
2. TE 2 - Terminal Equipment 2 is a Non ISDN Device. (Analog Phone, PC - Port)
3. TA - Terminal Adapter converts TE2 into ISDN format.
4. NT2 - Network Termination 2 is a switching device.
5. NT 1 - Network Termination 1 connects equipment to ISDN structure.

R = Reference point between NonISDN TE2 and TA.

S = n n n Terminal TA and NT2.

T = n n n NT2 & NT1

U = n n n NT1 & ISDN Exchange.

ISDN Services -

→ Bearer Services

→ Tele services

→ Supplementary services

* Bearer services -

→ Pure transport of information between users. (layer 1-3)

→ Network does not change content.

→ Ex - Voice, data transfer.

* Tele services -

→ Provide complete communication. (layers 1-4)

→ Network + standardised application.

→ Ex - telephone, fax.

* Supplementary -

→ Additional feature.

→ Exchange a bearer - or tele service,

→ Ex - Caller ID, Call waiting.

D-Channel protocol-

<u>OSI Layers</u>	<u>Protocol</u>	<u>Function</u>
Layer 3	Q.931	Call setup, Connection, release.
Layer 2	Q.921	Framming and Error control.
Layer 1	1.430 (BRI)/ 1.431 PRI	Physical transmission, Line coding, timing, Power.

ISDN Engineering Success -

1. End to End digital communication.
2. Fast call setup.
3. Better voice quality.
4. Supports video, data, fax
5. Low noise affected and Low error rate.

Market Failure -

1. Its BRI speed was only 128 kbps.
2. More expensive.
3. Complex installation.
4. Replaced by DSL, Internet cable.
5. Circuit switch communication.

Telecom Final

OSI Model

OSI = Open System Interconnection.

Model Architecture -

1. Application

2. Presentation

3. Session

4. Transport

5. Network

6. Data link

7. Physical layer

Software / Upper layer

Hardware / Lower layer

* Objective -

Standardizing communication between disparate systems.

* OSI Model -

OSI model is a 7 layer reference model that explains how data travels from one to another computer through a network.

* Upper layer -

Handle communication services for applications.

* Lower layer -

Handle actual data transmission through the network.

Layer - 1 : Application layer

This is the closest layer to the user.

→ Identifying Partners -

Locating the destination system and confirming readiness.

→ Resource Availability -

Ensuring the target network device can accept the request.

→ Electronic messaging -

Providing the actual service for user and software network access.

Ex- HTTP, FTP, SMTP

Layer - 2 : Presentation Translation

It means the translation layer of OSI model.

Raw data → Translation → Data Compression → Data Encryption.

→ Translation -

Converts data formats to binary. (ASCII → EBCDIC)

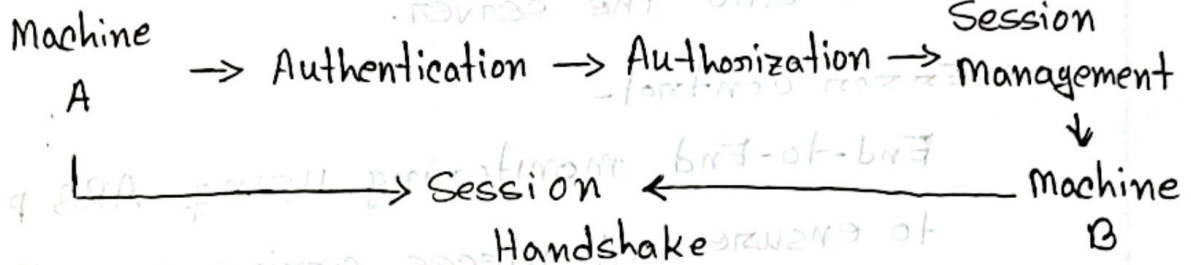
→ Data Compression -

Reduces file size to speed up transmission.

→ Data Encryption - Maintains data security for transit.

Ex- ASCII, EBCDIC, AFP.

Layer 3: Session Synchronizer



→ Authentication -

Verifying identity of any user.

→ Authorization -

Granting permission to access specific resources.

→ Session Management -

Establishes and manage the connection to maintain synchronization.

Ex- ADSP, ZIP, NetBIOS.

Layer 4: Transport Splitter - $\rightarrow$ Data from

Upper layer $\rightarrow$ Smaller units! $\rightarrow$ Data from

$\rightarrow$ Flow Control -

Manages the specific bandwidth between the user and the server.

$\rightarrow$ Error Control -

End-to-End monitoring using ARQ protocols to ensure all pieces arrive correctly.

Ex - TCP, UDP, SCTP.

Segments $\leftarrow$ Message



Seq. Number + Port number + Data Unit.

Layer 5: Network Router -



→ Path determination -

Creates the best route from sender to receiver.

→ Logical Addressing -

Assigning unique network identities using IPv4 or IPv6.

→ Subnet Operation -

Controls communication between different subnets.

Ex- IP Protocol, NAT, CIDR.

Segment

↓

Packet structure

Packet = Source IP + Destination IP + Data Unit + Segment.

Layer 6: Data Link Framing



→ Media access control (MAC) -

Manages how devices share the physical medium.

→ Physical addressing -

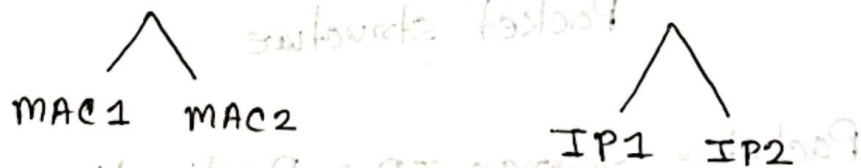
Utilizes manufacturer embedded MAC addresses.

→ Error control -

Detects errors occurring over a single network link.

Ex - Stop and Wait, ISDN, ARP.

Frame = MAC Source/Dest + IP Source/Dest + Segment



Layer 7 : Physical wire

Known as lowest layer of OSI model.

Transmitting raw bits over the communication channel.

Data units bit $\rightarrow$ 0 & 1.

Addressing Summary -

<u>Address Type</u>	<u>Layer</u>	<u>Purpose</u>
1. PORT Address	4	Identifies the ^{Application.} location.
2. IP Address	5	* Header of the segment. Identifies the device on the network.
3. MAC Address	6	* Header of the packet. Identifies the physical network interface.
		* Header of the Frame.

<u># Layer</u>	<u>Data Unit</u>
1, 2, 3	Data
Transport	Segment
Network	Packet
Datalink	Frame
Physical	Bits

TCP vs UDP :

Feature

TCP

UDP

Basic —

Transmission Control
Protocol

User Datagram
Protocol

Connection —

Connection-oriented

Connectionless

Reliability —

Reliable

Unreliable

Flow control —

Yes

No

Data Delivery —

Guaranteed

Not Guaranteed

Header size —

20-60 Bytes

8 Bytes

Packet order —

Maintains Order

Order not
Guaranteed.

Communication
Type —

Stream Oriented

Dataframe
Oriented

Example —

HTTP, FTP

Video streaming
Online games.

Ans 5

IPv4 vs IPv6 -

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<u>Feature</u>	<u>IPv4</u>	<u>IPv6</u>
Basic -	Internet Protocol version 4	Internet Protocol Version 6
Address length -	32 bits	128 bits
Address format -	Decimal	Hexadecimal
Number of address -	About 4.3 billion	about 3.4×10^{38}
Header size -	20 bytes	40 bytes
Complexity -	More Complex	Simpler and more efficient.
Security -	Optional	Built in support
Broadcast -	Supported	Not supported.

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